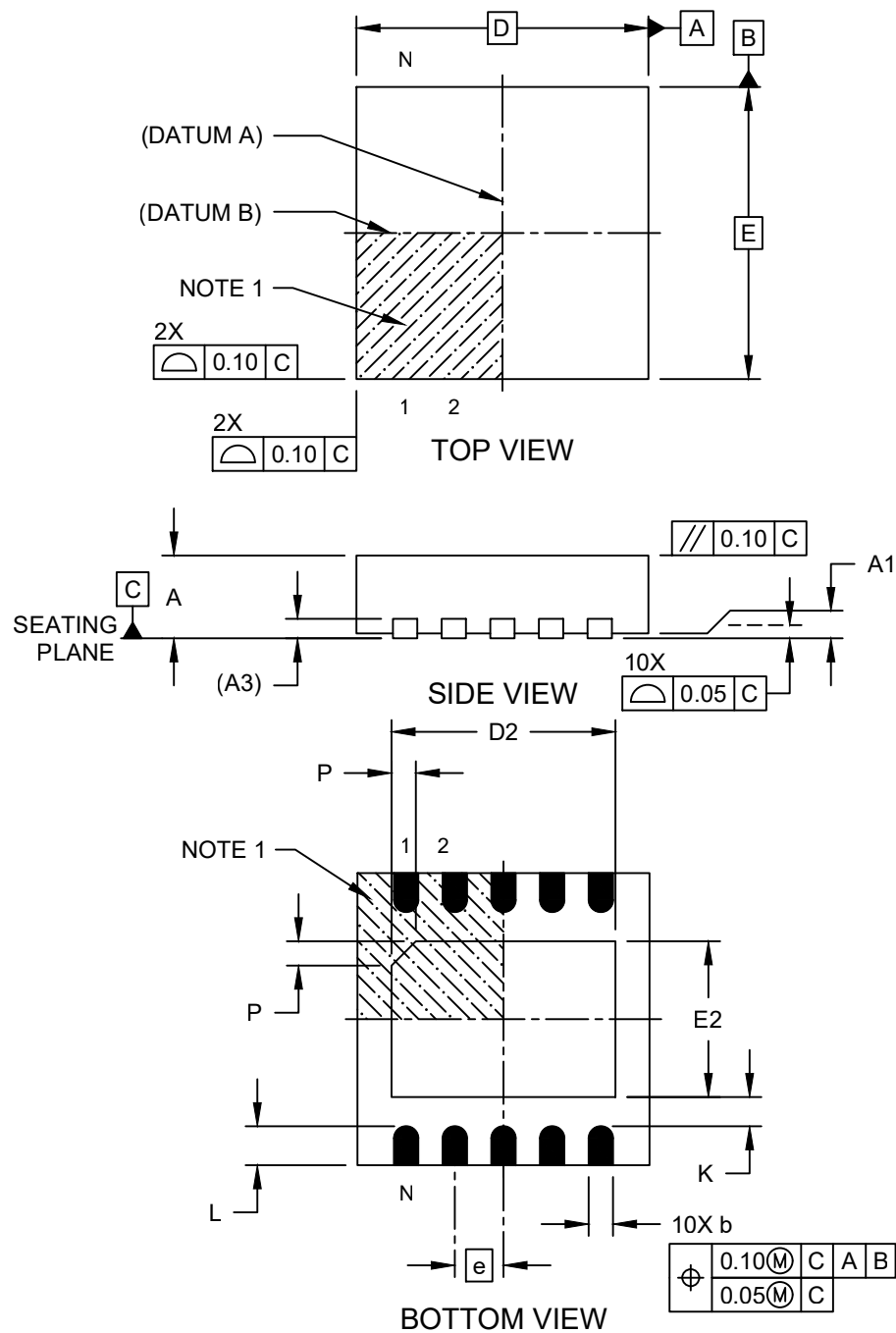


Packaging Diagrams and Parameters

10-Lead Very Thin Plastic Dual Flat, No Lead Package (9Q) - 3x3 mm Body [VDFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

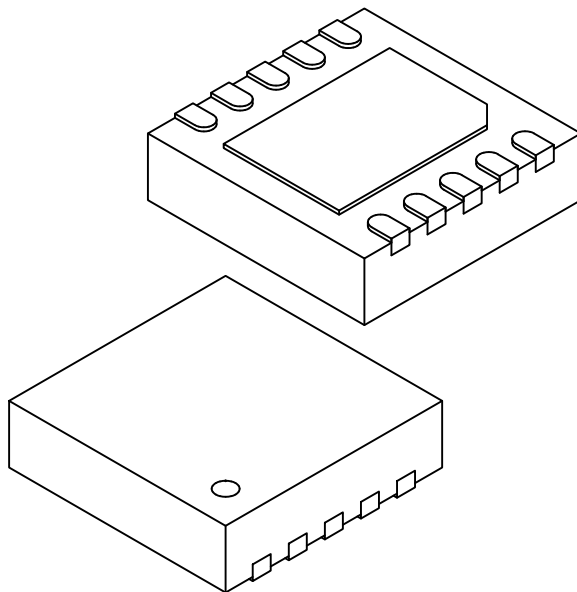


Microchip Technology Drawing C04-00206 Rev D Sheet 1 of 2

Packaging Diagrams and Parameters

10-Lead Very Thin Plastic Dual Flat, No Lead Package (9Q) - 3x3 mm Body [VDFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		10		
Pitch	e		0.50 BSC		
Overall Height	A		0.80	0.90	1.00
Standoff	A1		0.00	0.02	0.05
Terminal Thickness	(A3)		0.20 REF		
Overall Length	D		3.00 BSC		
Exposed Pad Length	D2		2.20	2.30	2.40
Overall Width	E		3.00 BSC		
Exposed Pad Width	E2		1.50	1.60	1.70
Exposed Pad Chamfer	P		-	0.25	-
Terminal Width	b		0.18	0.25	0.30
Terminal Length	L		0.35	0.40	0.45
Terminal-to-Exposed-Pad	K		0.25	0.30	-

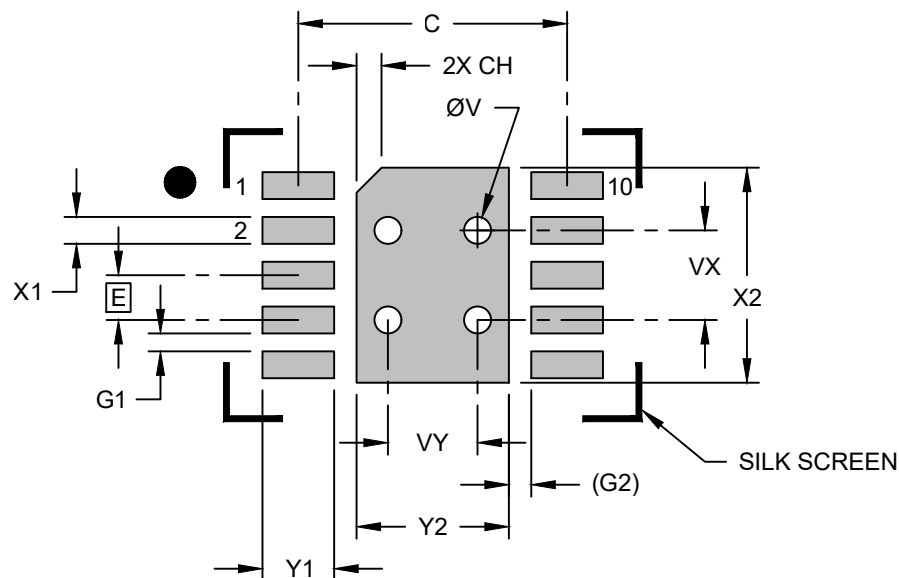
Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

10-Lead Very Thin Plastic Dual Flat, No Lead Package (9Q) - 3x3 mm Body [VDFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Optional Center Pad Width	Y2			1.70
Optional Center Pad Length	X2			2.40
Contact Pad Spacing	C		3.00	
Center Pad Chamfer	CH		0.28	
Contact Pad Width (X10)	X1			0.30
Contact Pad Length (X10)	Y1			0.80
Contact Pad to Contact Pad (X8)	G1	0.20		
Contact Pad to Center Pad (X10)	G2	0.25 REF		
Thermal Via Diameter	V		0.30	
Thermal Via Pitch	VX		1.00	
Thermal Via Pitch	VY		1.00	

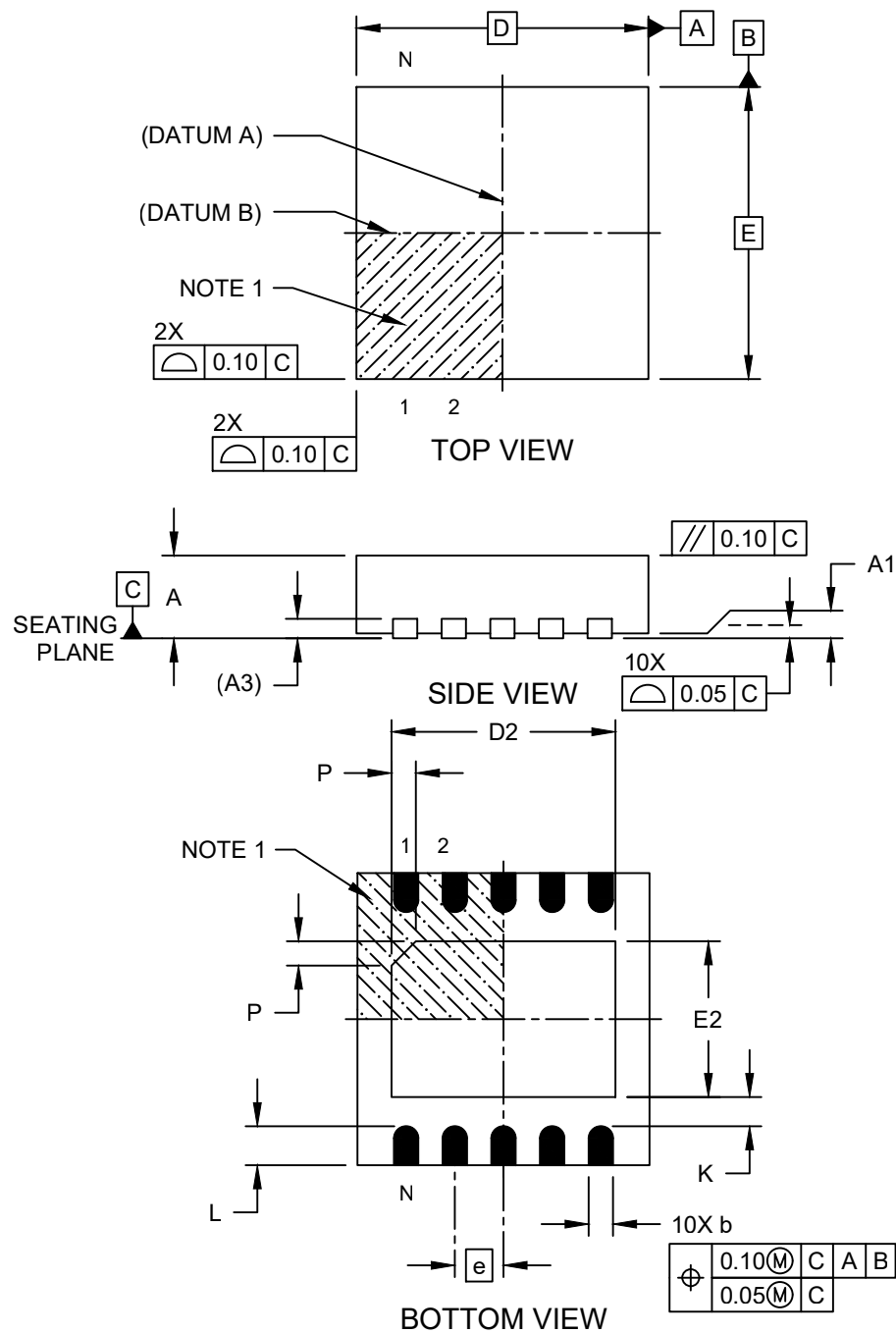
Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerances, for reference only.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Packaging Diagrams and Parameters

10-Lead Very Thin Plastic Dual Flat, No Lead Package (9QX) - 3x3 mm Body [VDFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

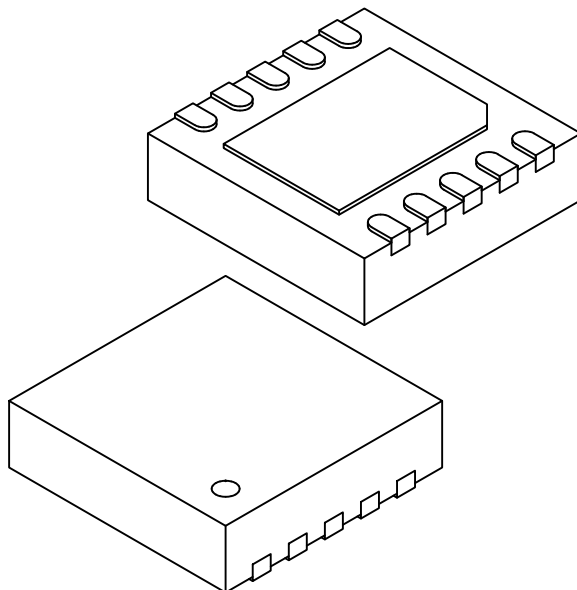


Microchip Technology Drawing C04-00206 Rev D Sheet 1 of 2

Packaging Diagrams and Parameters

10-Lead Very Thin Plastic Dual Flat, No Lead Package (9QX) - 3x3 mm Body [VDFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		10		
Pitch	e		0.50 BSC		
Overall Height	A		0.80	0.90	1.00
Standoff	A1		0.00	0.02	0.05
Terminal Thickness	(A3)		0.20 REF		
Overall Length	D		3.00 BSC		
Exposed Pad Length	D2		2.20	2.30	2.40
Overall Width	E		3.00 BSC		
Exposed Pad Width	E2		1.50	1.60	1.70
Exposed Pad Chamfer	P		-	0.25	-
Terminal Width	b		0.18	0.25	0.30
Terminal Length	L		0.35	0.40	0.45
Terminal-to-Exposed-Pad	K		0.25	0.30	-

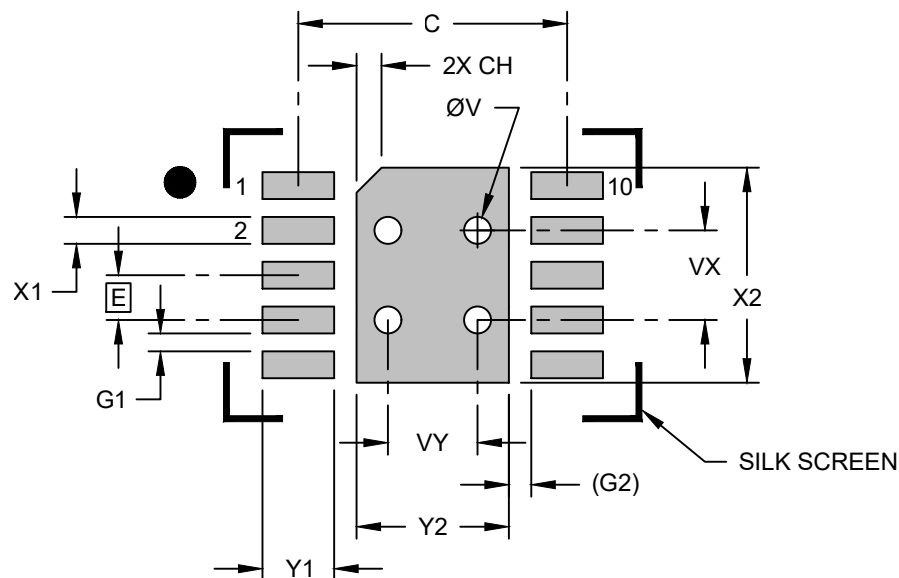
Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

10-Lead Very Thin Plastic Dual Flat, No Lead Package (9QX) - 3x3 mm Body [VDFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Optional Center Pad Width	Y2			1.70
Optional Center Pad Length	X2			2.40
Contact Pad Spacing	C		3.00	
Center Pad Chamfer	CH		0.28	
Contact Pad Width (X10)	X1			0.30
Contact Pad Length (X10)	Y1			0.80
Contact Pad to Contact Pad (X8)	G1	0.20		
Contact Pad to Center Pad (X10)	G2	0.25 REF		
Thermal Via Diameter	V		0.30	
Thermal Via Pitch	VX		1.00	
Thermal Via Pitch	VY		1.00	

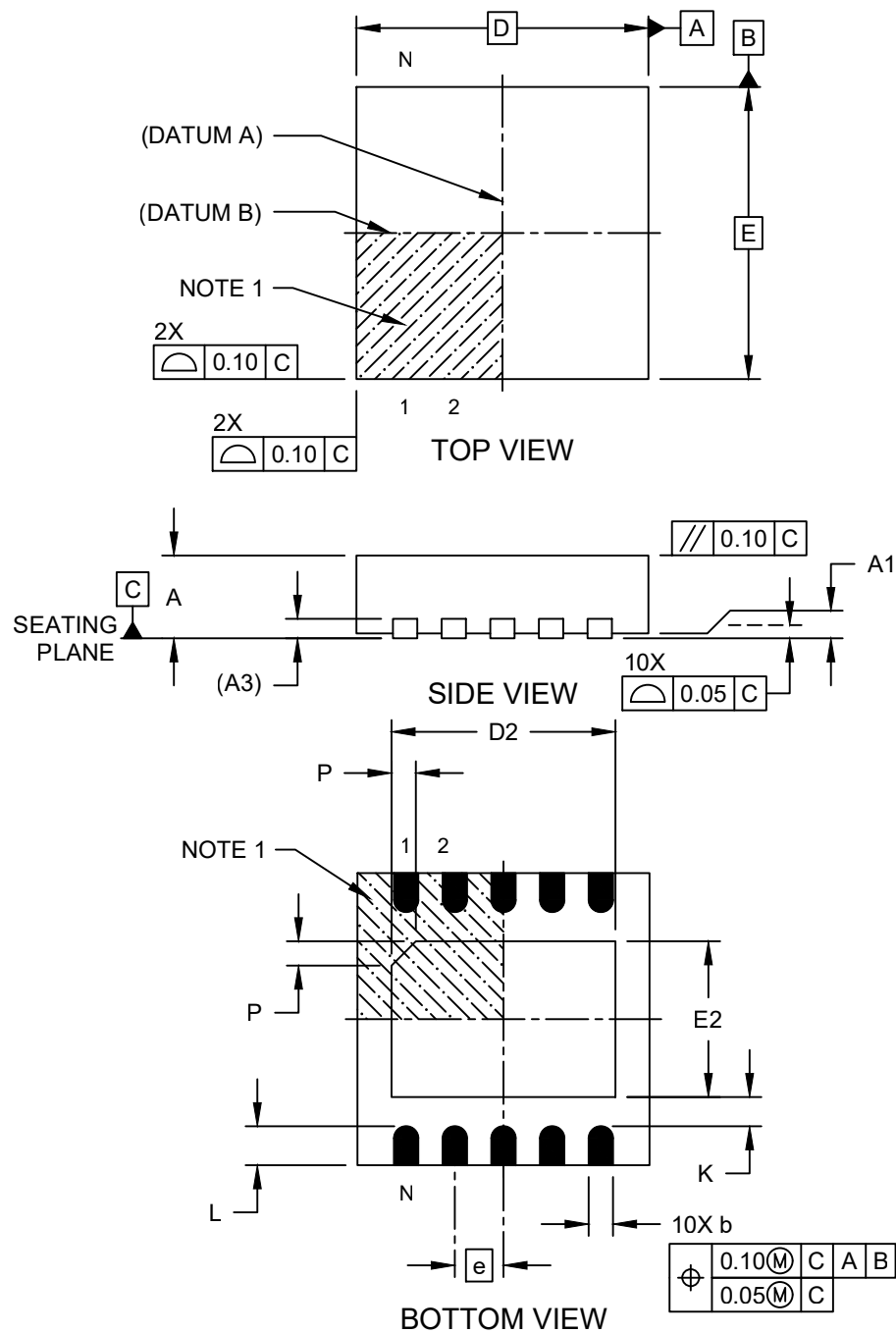
Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerances, for reference only.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Packaging Diagrams and Parameters

10-Lead Very Thin Plastic Dual Flat, No Lead Package (QSN) - 3x3 mm Body [VDFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>

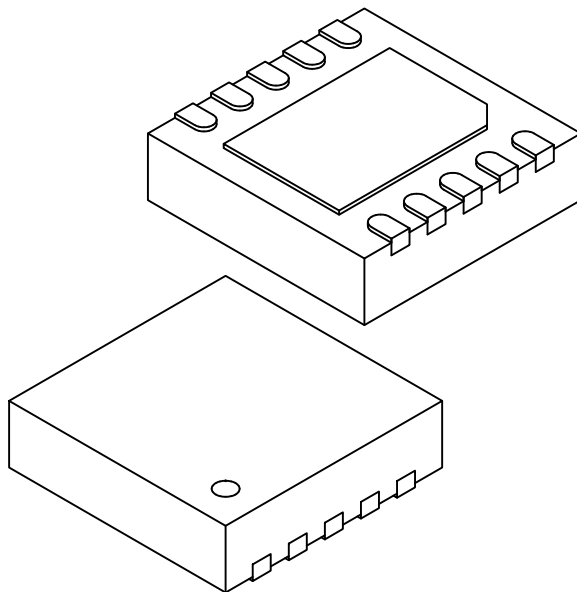


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Packaging Diagrams and Parameters

10-Lead Very Thin Plastic Dual Flat, No Lead Package (QSN) - 3x3 mm Body [VDFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



		Units	MILLIMETERS		
Dimension Limits			MIN	NOM	MAX
Number of Terminals	N		10		
Pitch	e		0.50 BSC		
Overall Height	A		0.80	0.90	1.00
Standoff	A1		0.00	0.02	0.05
Terminal Thickness	(A3)		0.20 REF		
Overall Length	D		3.00 BSC		
Exposed Pad Length	D2		2.20	2.30	2.40
Overall Width	E		3.00 BSC		
Exposed Pad Width	E2		1.50	1.60	1.70
Exposed Pad Chamfer	P		-	0.25	-
Terminal Width	b		0.18	0.25	0.30
Terminal Length	L		0.35	0.40	0.45
Terminal-to-Exposed-Pad	K		0.25	0.30	-

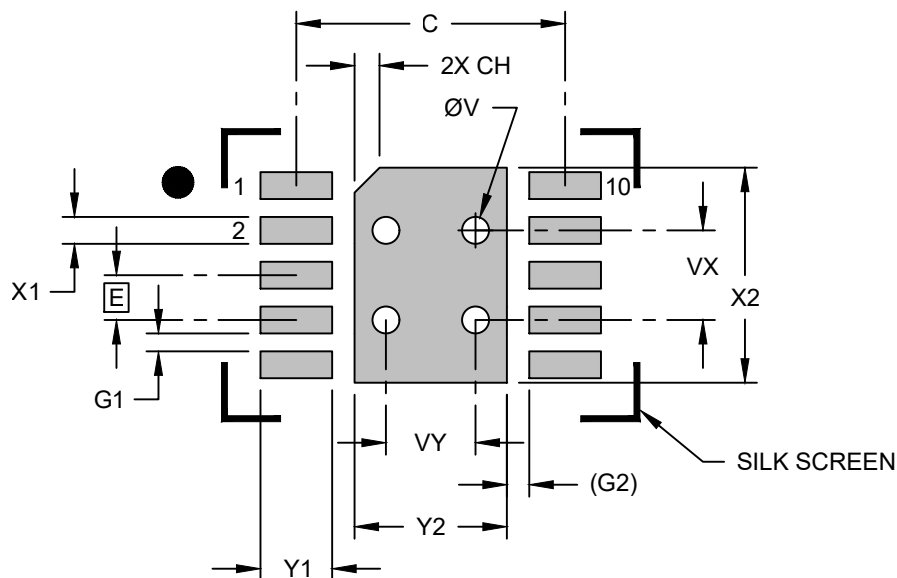
Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

Land Pattern

10-Lead Very Thin Plastic Dual Flat, No Lead Package (QSN) - 3x3 mm Body [VDFN]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

	Units	MILLIMETERS		
		MIN	NOM	MAX
Contact Pitch	E	0.50 BSC		
Optional Center Pad Width	Y2			1.70
Optional Center Pad Length	X2			2.40
Contact Pad Spacing	C		3.00	
Center Pad Chamfer	CH		0.28	
Contact Pad Width (X10)	X1			0.30
Contact Pad Length (X10)	Y1			0.80
Contact Pad to Contact Pad (X8)	G1	0.20		
Contact Pad to Center Pad (X10)	G2	0.25 REF		
Thermal Via Diameter	V		0.30	
Thermal Via Pitch	VX		1.00	
Thermal Via Pitch	VY		1.00	

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerances, for reference only.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process